

Zhengran Ji

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Website: <https://jzr01.github.io/>

Education

Duke University, Durham, NC

May 2025

Masters of Computer Science (Research Track)

Thesis: *Strategic Agent Learning with Human-in-the-Loop*

Advisor: Boyuan Chen

University of California, Irvine, Irvine, CA

June 2023

Bachelor of Science in Data Science (Honors Program)

Bachelor of Science in Mathematics (Honors Program)

Phi Beta Kappa, Cum Laude, Pi Mu Epsilon

Publications

- [1] Lingyu Zhang, **Zhengran Ji**, Nicholas R. Waytowich, Boyuan Chen, *GUIDE: Real-Time Human-Shaped Agents*, **NeurIPS 2024**, Sep 2024.
- [2] **Zhengran Ji**, Lingyu Zhang, Paul Sajda, Boyuan Chen, *HUMAC: Enabling Multi-Robot Collaboration from Single-Human Guidance*, arXiv, Sep 2024.
- [3] Lingyu Zhang, **Zhengran Ji**, Boyuan Chen, *CREW: Facilitating Human-AI Teaming Research*, arXiv, Jul 2024.
- [4] **Zhengran Ji**, Mike Hu, and Huolin Xin, *MnEdgeNet for Accurate Decomposition of Mixed Oxidation States for Mn XAS and EELS L_{2,3} Edges without Reference and Calibration*, **Scientific Report**, Aug 2023.
- [5] Huolin Xin, Chunyang Wang, **Zhengran Ji**, Mike Hu, and Lingli Kong, *A Universal Data Synthesizer to Enable AI4TEM, Microscopy and Microanalysis*, Jul 2023.
- [6] Huolin Xin, Chunyang Wang, Dong Zhu, **Zhengran Ji**, Mike Hu, and Lingli Kong, *Decoding Spatial Symmetry and EELS Spectroscopic Fine Structures*, **Microscopy and Microanalysis**, Jul 2023.
- [7] Lingli Kong, **Zhengran Ji**, and Huolin Xin, *Electron Energy Loss Spectroscopy Database Synthesis and Automation of Core-Loss Edge Recognition by Deep-Learning Neural Networks*, **Scientific Report**, Dec 2022.

Research Experience

Duke University General Robotics Laboratory

Sep 2023 – Present

- Training Effective Agent Policies with Human-in-the-Loop
- Human-Guided Reinforcement Learning, Multi-agent/multi-robot systems

UCI DeepEM Laboratory

Jun 2021 – Sep 2023

- Developing Intelligent Transmission Electron Microscope (TEM) using Deep Learning

Awards

UCI Chancellor's Award of Distinction

Apr 2023

UCI Chancellor's Award of Excellence in Research

May 2023

UCI 3rd ICS Expo 1st Place Prize in ICS Honors Program

May 2023

UCI Math Undergraduate Research & Departmental Service Award

May 2023

UCI Math Honors Award

May 2023

UCI Dean's List

Jan 2020 – Jun 2023

UCI Outstanding Service and Online Mentorship Award

Jun 2021

Fellowships

UCI Undergraduate Research Opportunity Program Fellowship

Teaching Experience

Duke Graduate Research Assistant

- General Robotics Laboratory Fall 2023 - Present

Duke Graduate Teaching Assistant

- COMPSCI 230: Discrete Mathematics for Computer Science Fall 2023

UCI Learning Academic Resource Center Tutorial Leader

- Math 2B: Single Variable Calculus II Sep 2022 – Jun 2023

UCI Undergraduate Teaching Assistant

- Math 2B: Single Variable Calculus II Spring 2023, Winter 2023, Spring 2022
- I&C ICS 46: Data Structure Implementation and Analysis Winter 2023
- Math 1B: Pre-Calculus II Winter 2023, Winter 2022
- Math 9: Introduction to Programming for Numerical Analysis Fall 2022
- Math 120A: Abstract Algebra Spring 2022

UCI Undergraduate Grader

- Math 130B: Probability II Spring 2022
- Math 130A: Probability I Winter 2022, Fall 2022

Peer Academic Advisor

- UCI Undergraduate/Undeclared Program Mar 2022 – Jun 2023

UCI Math Circle Mentor

Jan 2021 – Dec 2021